
Report on Supervised Independent Work

Subject: Laser cooling and trapping of Rubidium atoms

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1 Introduction

This report is the result of a week of practical work conducted at the European Center for Quantum Sciences (CESQ) on a Magneto-Optical Trap (MOT). During these experiments, we observed the setup of an MOT, conducted experiments on it, and attempted to optimize it. This report is divided into several sections. First, it will introduce the operating principle of an MOT, and finally, it will describe the work carried out during the week.

2 How a MOT works

A Magneto-Optical Trap (MOT) is a cornerstone of modern atomic physics. It is the primary tool used to cool neutral atoms from room temperature—where they zip around at hundreds of meters per second—down to temperatures just a fraction above absolute zero (typically in the microkelvin range). Tapping cold atoms is a key step for several applications. The MOT is essential to create atomic clocks, vital for GPS technology or to measure fundamental constants and testing the limits of the standard model of physics. It can also be used to create qubits : trapped cold atoms can be loaded into optical lattices.

2.1 Cooling atoms with laser

Before, trapping the atoms, they need to be cooled down.

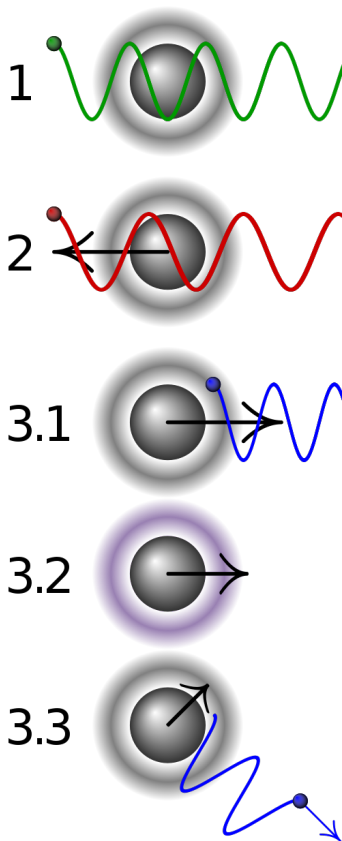
2.1.1 The Physics of Photon Momentum

Every photon carries a small amount of momentum, defined by the equation:

$$p = \frac{h}{\lambda}$$

Where h is Planck's constant and λ is the wavelength of the light. When an atom absorbs a photon, it must conserve momentum. If the photon is traveling in the opposite direction of the atom, the atom slows down. While a single "kick" is tiny (changing the atom's speed by only a few centimeters per second), an atom can undergo millions of absorption-emission cycles per second, leading to massive deceleration—stronger than the force of gravity.

2.1.2 Doppler Tuning



The real key of laser cooling is ensuring the atom only absorbs photons that are coming from the direction it is moving toward. This is achieved through the Doppler Effect. When the atom is going in the same direction as the photon (case 2 on Figure 1), its frequency is seen reddish by the atoms. Otherwise, when they are going in opposite direction, the photon frequency is seen bluish (case 3.1 on Figure 1). This is why the laser frequency is tuned (f_{laser}) to be slightly lower than the frequency the atom "wants" to absorb (f_0). An atom at rest sees the red-detuned light, perceives it as the "wrong" color, and ignores it. If an atom moves toward a laser beam, the light is Doppler-shifted to a higher frequency (bluer). This shift brings the light into perfect resonance with the atom. The atom "sees" the light coming at it as being the correct frequency to absorb, while the light chasing it from behind stays red-shifted even further away from resonance. Consequently, the atom is constantly being pushed against its direction of motion, no matter which way it turns.

Figure 1: Doppler cooling scheme

2.1.3 Optical Molasses

By using three pairs of counter-propagating beams (up/down, left/right, forward/back), the atom is trapped in a region where every movement is met with a resistive force. This force F for low velocities v can be approximated as:

$$F \approx -\beta v$$

Where β is a damping coefficient. Because this force behaves exactly like a particle moving through a thick syrup, this configuration is famously called Optical Molasses.

2.1.4 The Spontaneous Emission Cycle

After an atom absorbs a photon and slows down, it is in an "excited state." To absorb another photon and slow down more, it must first return to the "ground state" by emitting a photon. **Random Direction:** The atom emits this photon in a random direction. **Net Zero Force:** Over thousands of cycles, the "kicks" from these random emissions average out to zero. **The Net Win:** Only the absorption kicks (which were all directed against the atom's motion) add up to a significant slowing force.

2.1.5 The Doppler Cooling Limit

You might think we could cool atoms to a complete standstill ($0K$), but there is a theoretical floor. Because the absorption and emission of photons is a "jittery" quantum process, it introduces a small amount of heat (recoil heating). The lowest temperature achievable through this standard method is the Doppler Limit, given by:

$$T_D = \frac{\hbar\gamma}{2k_B}$$

Where γ is the linewidth of the atomic transition. For Rubidium, a common atom used in these experiments, this limit is about 140 microkelvins.

2.2 Magnetic Trapping

For neutral atoms such as rubidium, Paul or Penning traps are not suitable. Magnetic confinement is therefore achieved through a quadrupole magnetic field, generated using Helmholtz coils in an anti-Helmholtz configuration (Figure 4) [1]. In this setup, a current is circulated in one direction in one coil, and in the opposite direction in the other.

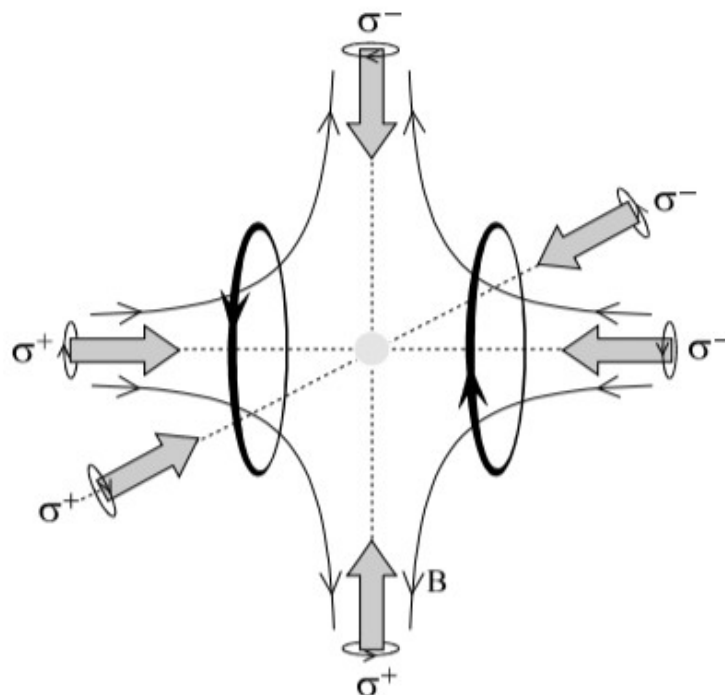


Figure 2: Anti-Helmholtz coil configuration for magnetic trapping.

At its center, there is a zero-field region, which allows for the confinement of rubidium atoms.

Outside of this neutral zone, there is a magnetic field gradient that is useful for laser cooling. This is because the magnetic field induces the lifting of energy level degeneracy, allowing us to exploit the hyperfine levels, a phenomenon known as the Zeeman effect (Figure 3) [2].

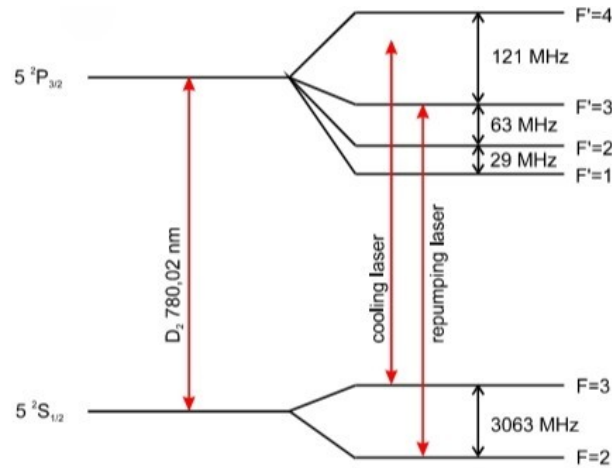


Figure 3: Zeeman effect on hyperfine levels.

3 Work completed during the week

3.1 Observational and literature review

On our first two days of this project, we first met our supervisor, Matthew White. He explained the objective of the project, what we would be doing, and what the challenges were.

We then met two students who were also working on the MOT. These students were in the process of finalizing the setup of the trap. As an introduction to this work, the students gave us a general overview of the MOT's implementation and answered all our questions. To delve deeper into the subject, we began conducting a literature review on the topic. We explored the theoretical principles governing magneto-optical traps, particularly laser cooling and magnetic field trapping.

Later, we observed the installation of the MOT with great curiosity. In particular, we observed certain technical requirements, such as the proper distribution of laser beams within the MOT and the correct adjustment of polarization. We also gained an understanding of certain aspects of the MOT's operation, which will be detailed later in this report. By the end of the two days, we were able to observe a fully functional MOT.

3.2 Characterization of the magneto-optical trap

From the third day onward, we were able to work with a fully operational magneto-optical trap. The startup procedure involved several sequential steps. First, the infrared laser used for the experiment was switched on and its operating parameters, such as temperature, were adjusted. The rubidium source was then activated by applying a current between 6 and 7 A, allowing the release of atomic vapor inside the chamber. A visualization software was launched to monitor both the laser spectrum and the trap using a camera. Finally, a current generator was turned on to supply the magnetic field coils.

A crucial step consisted in tuning the laser frequency using the spectral visualization software. We focused on a specific feature of the spectrum, corresponding to an absorption dip associated with an atomic transition of rubidium. This structure arises from the interaction between the laser and the hyperfine energy levels of the atoms, where only photons with a well-defined frequency can drive transitions between specific ground and excited states.

The laser was then frequency-locked to a precise point within this feature. This locking process consists in actively stabilizing the laser frequency on a chosen reference point of the atomic spectrum, using a feedback loop. Any deviation of the laser frequency from this set point is continuously corrected, ensuring long-term stability of the emission.

This stabilization is essential for laser cooling and trapping. In a MOT, the laser must be slightly red-detuned from the atomic resonance in order to create a velocity-dependent scattering force that slows down the atoms. Even small frequency drifts can shift the detuning away from its optimal value, reducing the cooling efficiency and potentially preventing the formation of a stable trap.

The locking therefore guarantees both short-term and long-term stability of the experimental conditions. In practice, this stability was directly observable on the spectral monitor as a steady signal over time, without noticeable drift of the locking point. Experimentally, this translated into a more robust and reproducible MOT, where the trapped atomic cloud remained stable in position and intensity over extended periods.

Once all these steps were completed, we could observe a stable laser signal and the formation of the MOT on the camera, appearing as a bright spot at the intersection of the laser beams. As we can see in the figure 4. The bright spot observed on the camera corresponds to the fluorescence of the trapped rubidium atoms.

We then proceeded to characterize the MOT by studying the influence of several experimental parameters. Starting from a set of parameters yielding a stable trap (laser temperature of 30.54°C, setpoint temperature of 30.52°C, laser current of 102.00 mA, repump frequency of 2950 MHz, and coil current of 1.20 A), we varied one parameter at a time while keeping the others constant.

The variation of the laser temperature showed a strong impact on both the stability and visibility of the MOT. A slight increase in temperature led to a decrease in intensity and stability, and beyond a certain threshold, the laser could no longer be locked, resulting in the disappearance of the trap. This highlights the sensitivity of the system to

the laser frequency, which is directly linked to the diode temperature.

The coil current also played a key role. We observed that the size of the trapped atomic cloud decreased as the current was reduced, indicating that the magnetic field gradient directly affects the spatial confinement of the atoms. Below a certain threshold, the MOT disappeared entirely.

Finally, variations in the laser current influenced the conditions under which the MOT could form, notably by modifying the threshold coil current required for trapping.

Overall, these observations demonstrate that the MOT is highly sensitive to experimental parameters, and that achieving optimal trapping requires careful tuning of the laser frequency, intensity, and magnetic field.

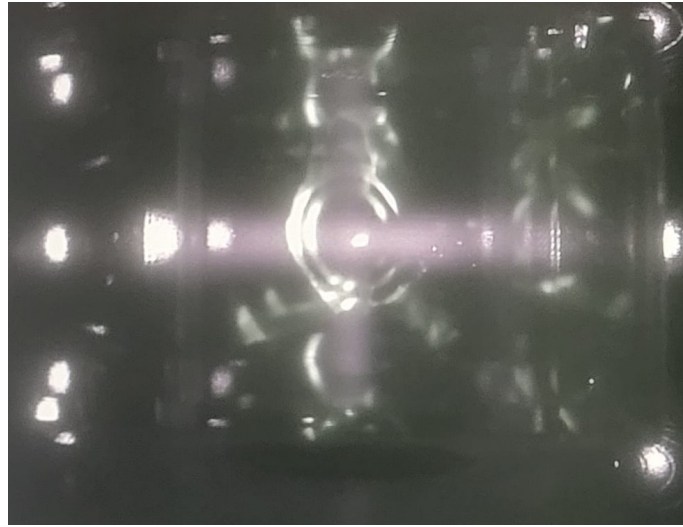


Figure 4: Camera image of an MOT

3.3 MOT optimization work

After becoming familiar with the operation of the MOT and performing initial characterizations, we were asked to slightly optimize the setup. One of the objectives was also to verify that the laser was addressing the correct atomic transition of rubidium. To achieve this, part of the laser beam had to be coupled into an optical fiber connected to another laboratory where further spectral analysis could be performed.

Before this, the optical fiber itself had to be made operational. The fiber provided by our supervisor had been heavily used and required refurbishment. Using a fiber polishing kit, we carefully polished the fiber end with successive polishing papers of decreasing grain size in order to restore a clean and usable surface.

Once the fiber was prepared, we modified the MOT optical setup to divert a portion of the laser into the fiber. To facilitate alignment, we first injected light into the fiber so that a visible beam emerged from its output. This reverse propagation allowed us to align the optical components more easily. The goal was to superimpose the beam exiting the fiber with the infrared laser beam used in the experiment, ensuring efficient coupling of the laser into the fiber. This alignment procedure proved to be particularly delicate and time-consuming.

The addition of new optical components required us to readjust the overall alignment of the setup, including mirrors and beam paths, as well as the distribution of optical power. In the MOT configuration, the beam is split such that approximately two-thirds of the optical power is directed toward the beams propagating in the horizontal plane (along the O_x and O_y axes), while the remaining one-third is used for the vertical axis (O_z). By carefully adjusting mirror

orientations and polarization-dependent elements, we were able to recover this required power distribution.

However, after these modifications, the MOT no longer formed: the characteristic bright spot at the center of the trap disappeared. This indicated that another crucial parameter had been affected, namely the polarization of the laser beams.

Polarization plays a key role in the operation of a MOT due to atomic selection rules. The trapping mechanism relies on circularly polarized light (σ_+ and σ_-), which drives specific transitions between magnetic sublevels of the atoms. In combination with the magnetic field gradient, this ensures that atoms preferentially absorb photons that push them back toward the center of the trap. If the polarization is not properly adjusted, these selection rules are no longer satisfied, and the restoring force responsible for trapping vanishes.

Before working on polarization, we first optimized the optical power of the system. By adjusting mirror orientations, we reached a maximum power of 18 mW in the relevant part of the setup.

We then focused on polarization optimization. A dedicated section of the optical setup allowed for fine control of the polarization state. We used a Polarization Analyzer (SK010PA) along with its associated software to monitor the polarization. The software displays the state of polarization on a Poincaré sphere, where each point represents a specific polarization state of the light. A stable polarization corresponds to a cluster of points localized in a small region, whereas fluctuations in polarization appear as a spread or drift of these points over time.

Additionally, when mechanical stress was applied to the optical fiber (for instance by slightly pressing on it), the polarization state evolved and traced a trajectory on the Poincaré sphere. This trajectory often takes the form of a circle, reflecting continuous changes in the relative phase between polarization components induced by birefringence in the fiber. Our objective was to minimize this effect, ensuring that the polarization remained as stable as possible.

This polarization optimization step was also time-consuming, requiring careful manual adjustments.

Finally, after completing these adjustments, we were able to recover a functioning MOT. However, due to time constraints, we were not able to push the optimization further or perform more advanced analysis

In parallel, we coded a program to calculate the size of the MOT (*main_sizeMOT*). To quantify the dimensions of the Magneto-Optical Trap (MOT), a custom image processing routine was developed using the OpenCV and NumPy libraries. The program first allows the user to define a Region of Interest (ROI) and establish a spatial scale by mapping a known physical distance to a pixel displacement. The MOT signal is isolated through dynamic thresholding based on an intensity percentile, followed by a contour detection algorithm that filters out background noise. By calculating the total number of pixels within the validated contours and applying the squared spatial scale factor, the software accurately determines the physical surface area of the trapped atomic cloud in real-world units .

Conclusion

This week allowed us to work directly on a device essential for quantum technologies and provided an introduction to laboratory work. We were able to discover concepts and techniques relevant to our professional future, such as the operation of a magneto-optical trap, mirror alignment, and all the procedures detailed in this report. Despite the short duration, we experienced both frustration and satisfaction when the magneto-optical trap finally worked. This week has increased our curiosity and our desire to work as close as possible to quantum technologies.

References

- [1] “Magneto-optical trap,” Wikipedia: The Free Encyclopedia, Wikimedia Foundation, 2026.
- [2] J. Zachorowski, T. Pałasz, and W. Gawlik, “Magneto-Optical Trap for Rubidium Atoms,” *Optica Applicata*, vol. 28, no. 3, p. 239, 1998.